

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-LV-6701-05 Sheet 1 of 1
OPL Title	Manual Bypass (HV-6701) Operation	Discipline	Process / Operations	
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	Area / Unit	6700 - LP SEPARATOR SYSTEM	
Related Interlock	SEQ-6701 (SIL 1)	P&ID Ref	TJC-LLD-PID-6701	
Classification:	☒ Basic Knowledge	☐ Improvement	☐ Trouble Case	Aspect: P-Q-C-D- S -M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **manual bypass (hv-6701) operation** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. The manual bypass lets operators keep the separator level safely while LV-6701 is isolated for maintenance. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Notify the panel and take LIC-6701 to manual	Permit valid & briefed
2	Crack open the bypass HV-6701 slowly	Parameter within limit
3	Throttle HV-6701 to hold FA-6710 level using LT-6710	Interlock status confirmed
4	Isolate LV-6701 (close block valves) for the work	Change logged on DCS
5	Reverse the steps and re-establish auto control afterwards	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 isolation for planned maintenance	Planned - bypass used to hold level	Lined up HV-6701 manual bypass, isolated LV-6701 safely
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Manual handwheel jack on LV-6701 seized	Corrosion in handwheel thread	Overhauled handwheel assembly, greased thread

6. KEY LEARNING POINTS

- The manual bypass lets operators keep the separator level safely while LV-6701 is isolated for maintenance.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

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